A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 18

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Number all possible sampling sites (eg on a grid map) **1** Select the sites using a random number generator / table – (regular shape: grid coordinates from random numbers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis **1** So the sample does not represent the whole area and conclusions are invalid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
02.1	**1 mark each** 1 mark for each of the 5 points below. **1 Food:** cod, tuna, swordfish, sharks, turtles – extinct - dodo, great auk, passenger pigeon **1 Fashion:** leopard, snow leopard, tiger, ocelot fur – fur seals – crocodile / alligator leather – kingfisher, parrot, ostrich feathers **1 Entertainment / pets:** parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants **1 Furniture / ornaments:** mahogany, teak, ramin timber – ebony and ivory piano keys – turtle shell, shark teeth, coral and shells as souvenirs **1 Traditional medicines:** tiger (bones, claws), rhinoceros horn, seahorses **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here.	5
02.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Threat to humans:** sharks, venomous snakes, crocodiles **1 Disease vectors:** malaria mosquitoes, tsetse flies **1 Predators of livestock:** wolves, puma, birds of prey, herons, seals **1** Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** High demand and prices make illegal trade / poaching profitable – laws are hard to enforce across borders **1** Belief in the medicine persists despite little scientific evidence – species become rarer and more valuable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps **1** Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** Fixed mooring buoys so boats do not anchor on coral **1** Education of divers / snorkellers not to touch coral or kick up sediment – bans on spear fishing and souvenir collection **1 Fishing controls:** no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling **1** Protection of turtle nesting beaches – control of coastal development – protection of mangroves and seagrass – artificial reefs (reef balls, sunk ships / oil rigs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** No sunlight so no symbiotic algae – polyps rely only on captured food so have far less energy and grow very slowly **1** Damage from trawls takes hundreds of years to repair – the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Deep-water trawling (also by-catch) – oil and gas exploration in deeper water – ocean acidification **1** Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	
04.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2** Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance – indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met **up to 2 Single species:** easy planting, thinning and felling over large areas – but each tree species supports its own community so few species overall **up to 2** Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them **up to 2 Simple age structure:** uniform size makes management easy but gives little variety of light, temperature, wind – no old trees or dead wood for invertebrates and woodpeckers **up to 2 Mixed woodland:** varied niches, stratification, dead wood, indigenous trees each with their own dependent species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline **1** Continuing to grow them ties up land and capital for little extra timber – natural forests keep old and dead trees **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp **1** Wood fuel replaced by fossil fuels / electricity – so traditional management declined and woods became uniform **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** The processes (eg regulation of atmospheric gases by photosynthesis and respiration) act in a dynamic equilibrium and cancel each other out **1** Because there is no overall change their actions go unnoticed / their importance is not appreciated until they are lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Invertebrates, fungi and bacteria break down / decompose dead organic matter, releasing nutrients for plant growth **1** The organic matter and humus produced hold the soil together (reducing erosion) and regulate the water cycle, giving more even river flow and less flooding **1** Organic acids from decomposition aid the weathering of rock, producing more soil and releasing more nutrients **1** Microbes carry out the processes of the carbon, nitrogen and phosphorus cycles so wastes do not build up and nutrients are not depleted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Trees reduce wind velocity so wind pollination would be ineffective / unreliable over long distances **1** Insects search over long distances for flowers so dispersed populations can be pollinated, and less pollen needs to be produced (energy saving) – specialised flowers reduce pollen being wasted on other species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**1 mark each** 1 mark for each of the 3 points below. **1 Dispersal is not haphazard:** animals live in the same habitat as the plant so are likely to carry seeds to locations where they can survive **1** Seeds can be carried longer distances than by wind and can be larger, containing more nutrients for the seedling **1** Seeds passing through the gut are deposited in faecal material which acts as a fertiliser, and passage through the intestine may stimulate germination **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump **1** About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages **1** Each age has its own community: newly cut areas get high light for wildflowers and butterflies – older areas give dense cover for nesting birds / dormice **1** Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour **1** Fewer young trees eaten so trees and riverside vegetation recovered **1** More trees allowed beaver populations to increase – beaver dams increased wetland area **1** Wetland species (birds, fish, invertebrates) became more common – overall biodiversity rose, though wolves cause problems for livestock farmers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species **1** Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	
07.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2.7 - 2.45 = 0.25 billion = 250 million years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (million years) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans **1** Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Oxygen in the atmosphere absorbed ultraviolet light **1** A dynamic equilibrium of reactions between O, O_2 and O_3 (the ozone layer) developed **1** Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) **1** Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved – many anaerobic Archaea and bacteria died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) **1** The lower CO_2 concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion **1** Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas **1** Regulation of atmospheric composition (eg oxygen / CO_2 by photosynthesis and respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Transpiration** → water lost as vapour from the leaves **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins) – wastes of one process are raw materials for another so nothing is depleted or accumulates **1** Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials **1 Human systems are linear:** extract - process - manufacture - use - discard – minerals are dispersed after use or mixed so recovery is difficult **1** Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up – fossil fuel use is inherently linear and unsustainable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life **1** They leach from landfill or are released by incineration – persistent, bioaccumulating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **incineration** → burning waste, usually with energy recovery • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
08.3	**1 mark each** 1 mark for each of the 4 points below. **1** Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) **1** Easy identification of materials (coding) and separation of components so they can be recycled **1** Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) **1** Use of recycled and recyclable / biodegradable materials – reduced packaging and material use (Cradle to Cradle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
09.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
09.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
10.1	1 mark for the point below. **1** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1

Do not write outside the box

Question	Marking guidance	Marks
10.2	**1 mark each** 1 mark for each of the 4 points below. **1** Negative feedback resists change / returns the system towards its previous state **1** eg increased temperature increases cloud cover and albedo, or higher CO_2 increases photosynthesis **1** Positive feedback increases the change / makes it self-sustaining **1** eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
10.3	1 mark for the point below. **1** Many species with a complex food web so the loss of one has a small effect – agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Principles of natural systems:** dynamic equilibria and negative feedback – diverse systems resistant to change – low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles) – low-temperature processes using enzymes – material cycles with wastes as raw materials – small number of abundant elements – biodegradable non-toxic wastes • **Human systems:** non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) • **Applications:** renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy • Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires) – tipping points • **Barriers:** population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure • Justified conclusion on how far natural principles can be applied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **sustainable** → meeting present needs without stopping future generations meeting theirs • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **biodegradable** → broken down by decomposers **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Natural systems have remained sustainable because they run on low energy-density renewable energy, mainly sunlight driving the hydrological, carbon and nitrogen cycles; their reactions occur at low temperatures because enzymes lower activation energies; materials cycle so the waste of one process is the raw material of the next; only a few abundant elements are used; and wastes are biodegradable and non-toxic. Negative feedback keeps them in dynamic equilibrium and high diversity makes them resistant to change.	
	Human systems contrast on every point: non-renewable fossil fuels, high-temperature processes such as the Haber process and incineration, linear use that disperses minerals and mixes materials that cannot be separated, and persistent toxins like PCBs and the metals in electronics. We also override negative feedback - when degraded soil reduces yields we add fertiliser, irrigation and pesticides - and trigger positive feedback such as permafrost melting.	
	Copying natural principles is already possible in many areas: renewable energy and heat pumps, bioleaching at low temperature, recycling and design for disassembly, industrial symbiosis, legumes and organic matter instead of manufactured nitrate, and retaining hedgerows for pest predators - the basis of the circular economy. The limits are population size and affluence, the attitude of urban consumers who never see depleted resources, cost, and the need for international cooperation. So the principles can make individual activities far more sustainable, but only a large-scale, cooperative shift in lifestyle would make the whole human system sustainable.	
	Weaker answer - Level 2 (5 marks)	
	Natural systems use renewable energy from the Sun and recycle materials, so nothing runs out and there is no waste. Humans use fossil fuels which will run out and produce waste that is toxic. Natural processes happen at low temperatures but the Haber process needs high temperatures. Humans could copy natural systems by using renewable energy like solar and wind, recycling more and using less fertiliser. This would make human activities more sustainable. However it would be expensive and people would have to change their lifestyle.	
	What makes the difference	
	• Level 2 states the contrast correctly but does not explain the mechanisms (enzymes, negative feedback, diversity, dispersal of minerals) that Level 3 uses.	
	• Level 3 applies the principles to specific technologies (bioleaching, heat pumps, industrial symbiosis, legumes); Level 2 stays at 'recycle more'.	
	• 'Evaluate the extent' - Level 3 distinguishes individual activities from the whole system in its conclusion; Level 2's evaluation is a single sentence about cost.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Setting priorities:** IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • **Legislation and protocols:** Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • **Captive breeding and release:** selection, difficulties, techniques, release site criteria, hard / soft release, monitoring – seed banks • **Habitat conservation:** creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • **Evaluation:** examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers) – cost, politics, climate change – judgement on the most effective combination	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	
	Model answer - Level 5 (23 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Biodiversity is lost through direct exploitation, habitat destruction, introduced species, pollution and changes to abiotic factors, so its conservation needs methods that address each cause, at scales from a single species to international law. Their effectiveness varies with the threat, the species and the resources available.	
	Deciding what to conserve comes first. The IUCN Red List uses population size, area, decline rate and probability of extinction to prioritise species, and endemic, keystone, flagship and evolutionarily distinct species receive weight because their loss has wider consequences. This is effective at directing scarce resources, but it favours well-studied vertebrates over invertebrates and plants.	
	Legislation and agreements protect species and habitats. In the UK the Wildlife and Countryside Act, SSSIs, NNRs and Marine Conservation Zones, and in Europe SACs, SPAs and the Habitats Directive, restrict damaging activities; internationally CITES controls trade in threatened species, the IWC regulates whaling, the Ramsar Convention protects wetlands and the Rio Convention on Biological Diversity commits nations to conservation. Their effectiveness depends on enforcement: CITES has reduced trade in ivory and rhino horn but poaching continues where enforcement is weak; the IWC moratorium allowed some whale populations to recover but Japan and Norway continued whaling; designations protect against development but not against pollution or climate change crossing boundaries.	
	In-situ habitat conservation and management are the most fundamental methods because they keep species with the community and abiotic conditions they depend on. Management uses ecological understanding: maintaining plagioclimaxes such as chalk grassland and heathland by grazing or burning, coppicing and pollarding woodland, controlling water levels, light and grazing, designing reserves with the right area, shape, biological corridors and buffer zones, and creating new habitats such as wetlands in worked-out gravel pits. Case studies show it works - coppiced woodland restored dormice and butterflies, wetland creation increased bitterns - but reserves are often small and fragmented and need permanent funding.	
	Ex-situ methods rescue species for which in-situ protection has failed. Captive breeding faces problems of space, diet, breeding triggers, social groups and small gene pools, addressed with studbooks, artificial insemination, embryo transfer and cryopreservation; seed banks such as the Millennium Seed Bank store plant genetic diversity cheaply. The Arabian oryx, California condor and pool frog show captive breeding can bring species back, but only where protected habitat exists for hard or soft release with post-release monitoring, and it is expensive and works for few species.	
	Ecological monitoring - image databases, eDNA, tracking, satellites - measures whether any of this is working and has improved the targeting of effort. Overall, legislation is effective when enforced, habitat management is the most effective and widely applicable method, and captive breeding is a last resort that succeeds only in combination with habitat protection. The Living Planet Index continues to fall, however, which shows that conservation methods are winning individual battles while the drivers of loss - population, consumption, land-use change and climate change - remain unaddressed.	

Do not write outside the box

Question	Marking guidance	Marks
	What makes the difference	
	• The essay is structured by **type of method** and evaluates the effectiveness of each in turn, with named examples for both successes and failures.	
	• It draws on several specification areas (biodiversity, conservation legislation, habitat management, captive breeding, ecological processes, monitoring) - the synoptic breadth Level 5 requires.	
	• The conclusion ranks the methods and then steps back to the bigger picture (Living Planet Index, underlying drivers) - a supported judgement rather than a summary.	
	• A Level 3 essay would describe the same methods accurately but with fewer examples, little evaluation of enforcement or cost, and a conclusion that all methods are important.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Resources:** wood, fibres, oils, fuels, food – new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • **Biomimetics:** wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • **Medicines:** poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks – few species yet studied – physiological research (marsupials, hippo, sponges) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools and inbreeding • **Ecosystem services:** atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • **Evaluation:** economic vs intrinsic / moral arguments – services unnoticed because of dynamic equilibrium – unknown future value – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box